Highway Engineering

- INS3441 -

PROBLEM SESSION 5



Question – 1

A two way double lane highway has a design speed of 80 km/h. It is planned that to place two consecutive curves with enough space between them. For the first curve superelevation will be implemented without ramp and for the second one superelevation will be implemented with a 43.71 m ramp. In curves 56% of the centrifugal force will be compensated by superelevation. Some data about this highway and criteria is given below.

Given:

$\Delta_1 = (63.66)°$; $\Delta_2 = (31.83)°$; $\Phi_1 = (2+750)$; $\Gamma_1 = (2+950)$; $R_2 = 380$ m

Platform width	= 11 m
Crown slope	= %2
Max. superelevation	= %8
l *(vehicle length)*	= 12 m

$q_{theo} = 0.786*(V_p^2 / R)$

Jerk $(p') = V^3 / (46.7*R*L) < 0.6$ m/sn^3

Expansion value $(b_g) = (n*l^2 / 2*R) + (0.05*V_p / R)$

a) Select the clothoid with smallest possible parameter and draw constant centerline superelevation diagram and make necessary calculations.

b) Find the length and parameter of the clothoid on the second curve and check if it is proper.

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In order to calculate the clothoid components, the necessary parameters are; the superelevation value, radius of curve and the maximum allowable speed on the curve. If there is no speed limitation the maximum speed is equal to the project speed.

Calculation of Curve Radius:

$$D = \text{Ғ}_{(km)} - \text{Ф}_{(km)} = [\ 2 + 950\ ] - [\ 2 + 750\ ] = 200\ \text{m}$$

$$D = \frac{2\pi R\Delta}{360} \qquad \frac{2 * \pi * R * 63,66}{360} = 200 \qquad R = 180\ m$$

Calculation of Superelevation Value:

$$q_{pra} = 0,56 * q_{teo} = 0,56 * 0,786 * \frac{V_p^2}{R} = 0,56 * 0,786 * \frac{80^2}{180} = 15,65 \quad > 8$$

Speed limit is required!

Max superelevation will be used as 8%.

$$8 = 0,56 * 0,786 * \frac{V_{p,new}^2}{180} \rightarrow V_{p,new}^2 = 57.2\ \text{km/h}$$

Clothoid parameter (A), should satisfy the 3 conditions and be checked for 3 controls.

Optical condition

$$\frac{R}{3} \le A \le R \quad \Rightarrow 60 \le A \le 180$$

Dynamic condition

$$A_{\min} = 0,19 * \sqrt{V_p^3} \quad = 0,19 * \sqrt{57,2^3} = 82,20$$

Superelevation condition

$$A_{min} = \sqrt{\frac{b}{2} \times \frac{(q_{min} + q_{max})}{0,005} \times R}$$

In the questions if it is not given as high standard road type do not check the superelevation condition.

Minimum length control (Lg = at least 45 m)

$$L_g = \frac{A^2}{R} \Rightarrow L_{g,\min} = \frac{A_{\min}^2}{R} = 45 \qquad 45 = \frac{A_{\min}^2}{180} \Rightarrow A_{\min} = 90 \qquad L_g = 45m$$

It is asked for minimum A parameter, so A is 90m. Now check for the other controls.

Important Note: «A» must be a multiple of 5!

Geometrical control

$$\frac{L_g}{2} \le \frac{D}{2} \quad \rightarrow \frac{45}{2} \le \frac{200}{2}$$

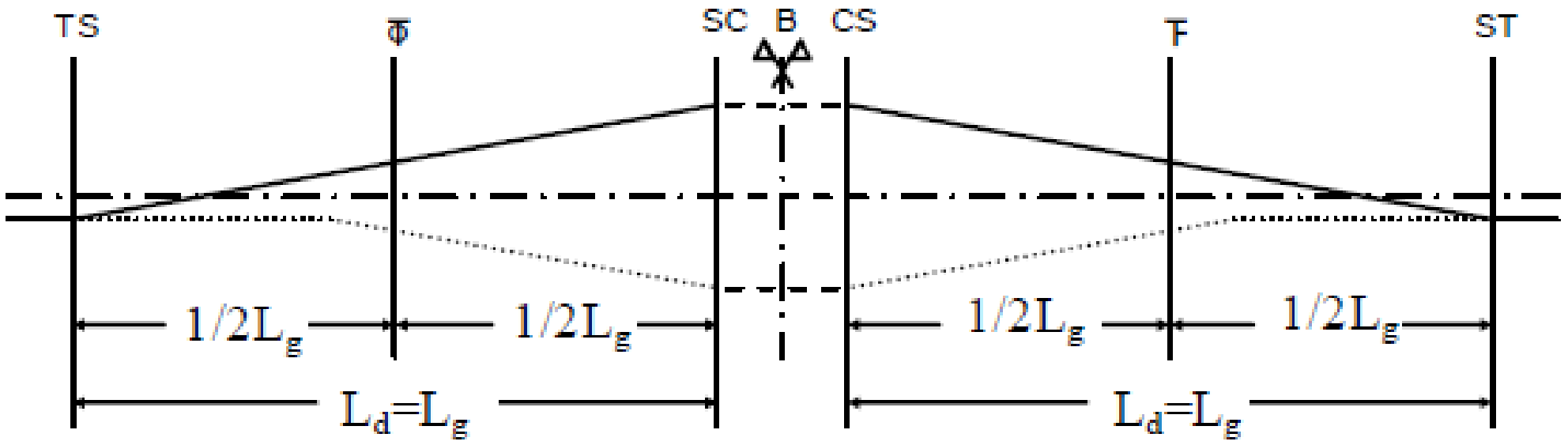
Jerk control

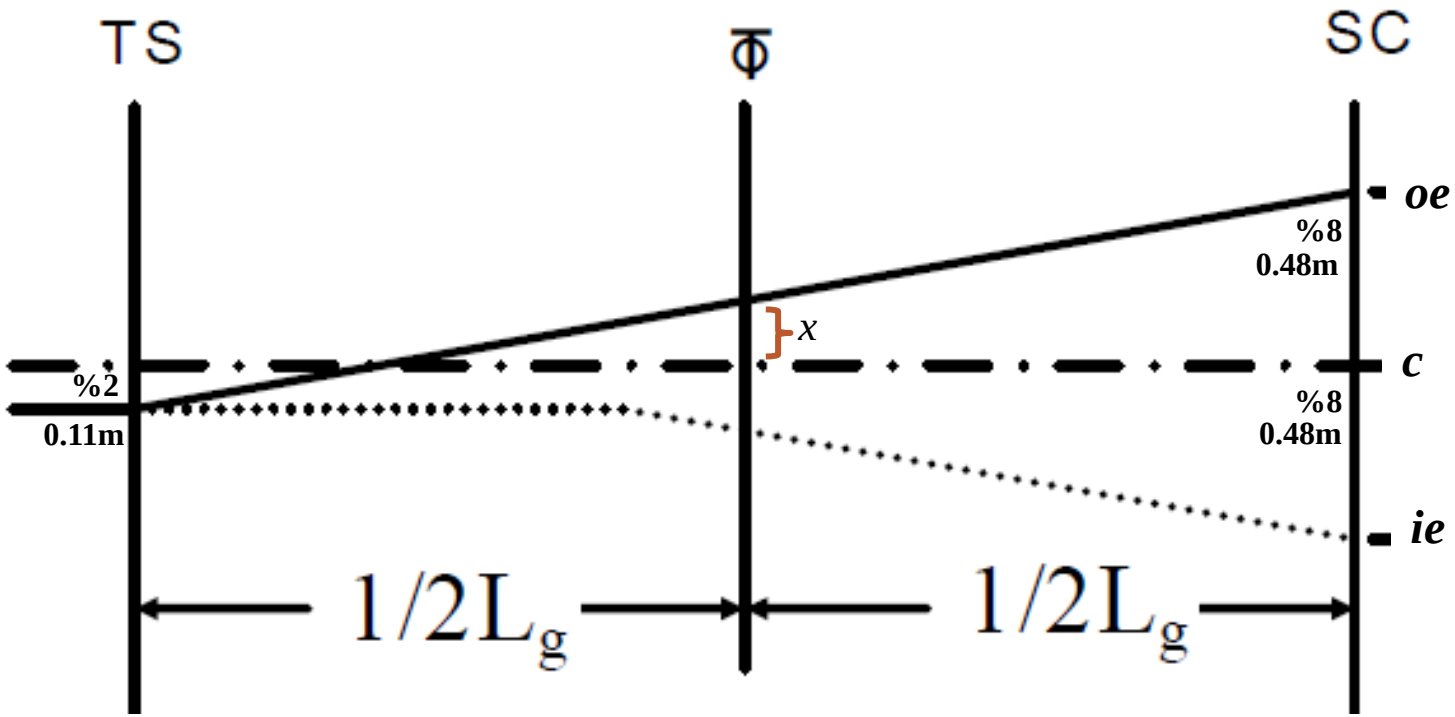
$$p' = \frac{V_p^3}{46,7 * R * L_g} = \frac{57,2^3}{46,7 * 180 * 45} = 0,49 m/s^3 \quad < 0,6 m/s^3$$

In order to draw the superelevation diagram platform width is required.

Since radius R is smaller than 200 m there should be expansion.

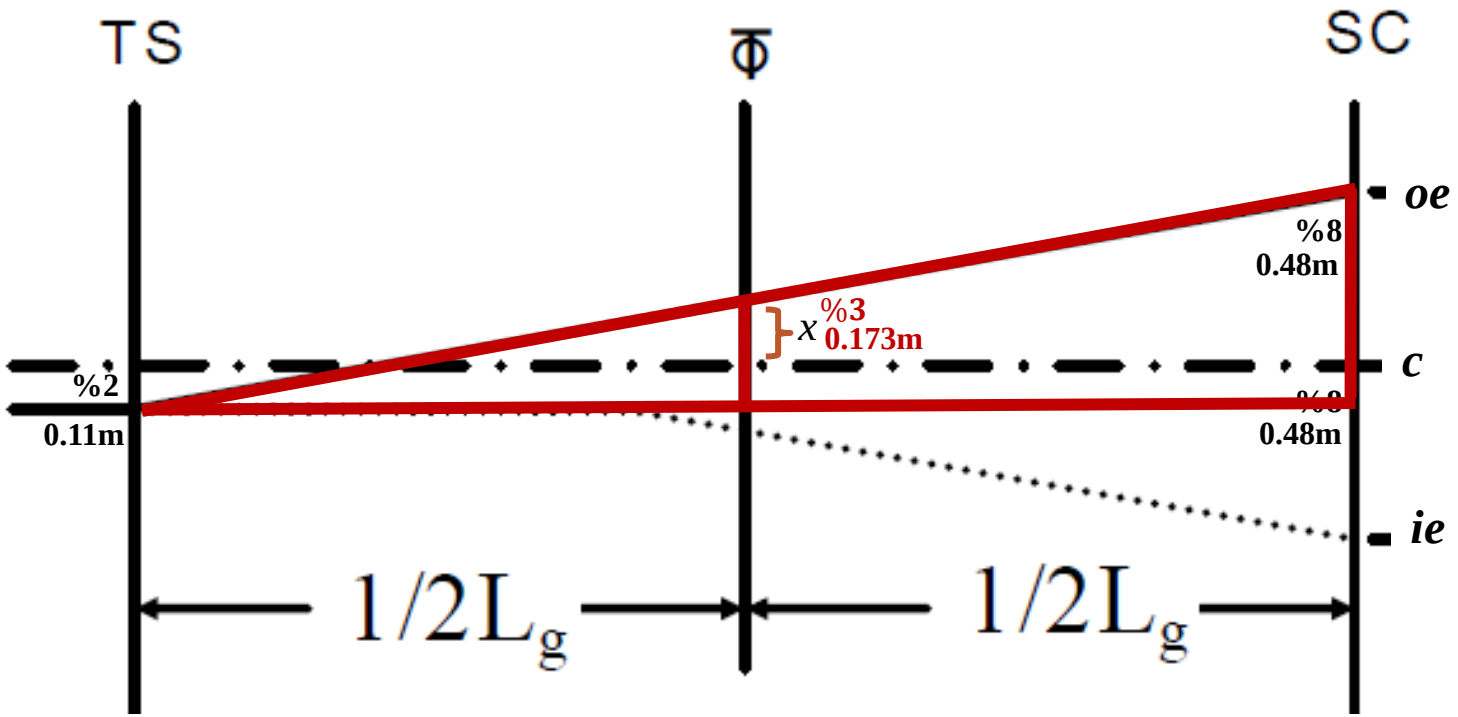
$$b_g = \frac{n*l^2}{2*R} + \frac{0,05V_p}{\sqrt{R}} = \frac{2*12^2}{2*180} + \frac{0,05*57,2}{\sqrt{180}} = 1,01m$$





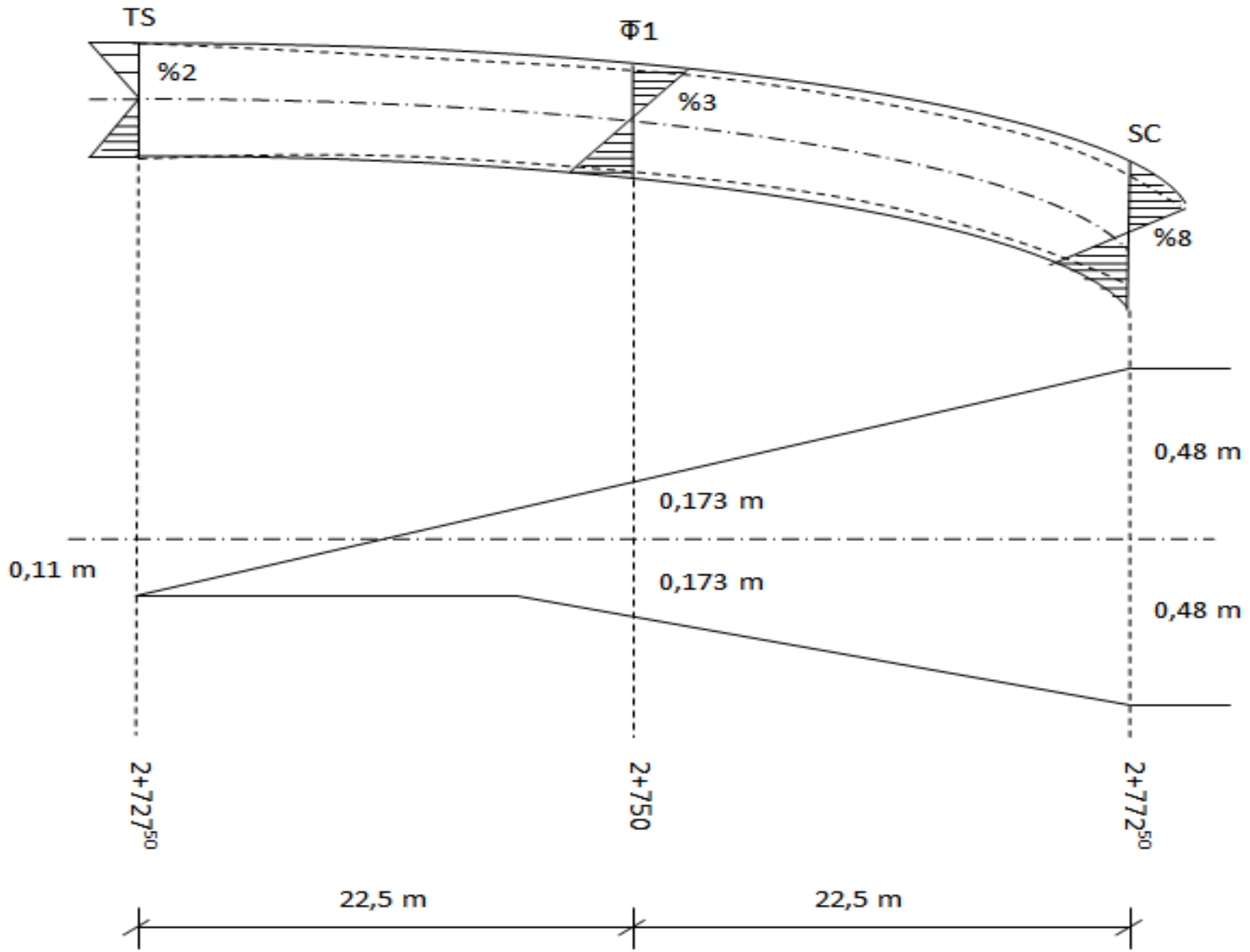
$$h_{TS} = q * \frac{b}{2} = \ 0,02 * \frac{11}{2} = \ 0,11m$$

$$h_{SC} = q * \frac{b}{2} = \ 0,08 * \frac{11+1,01}{2} = \ 0,48m$$



$$q_{\bar{\Phi}} \rightarrow \frac{x+2}{8+2} = \frac{1}{2} \qquad \Rightarrow x = \%3$$

$$h_{\bar{\Phi}} \rightarrow 0.03\,\frac{11+0.505}{2} = 0.173\ m$$


TS
Φ1
SC
%2
%3
%8
0,48 m
0,173 m
0,11 m
0,173 m
0,48 m
2+727,50
2+750
2+772,50
22,5 m
22,5 m

b) Find the length and parameter of the clothoid on the second curve and check if it is proper.

Ramp length formula:

$$K = \frac{2 * h_1 * L_g}{h_2 - h_1}$$

$$q_{pra} = 0,56 * q_{teo} = 0,56 * 0,786 * \frac{V_p^2}{R} = 0,56 * 0,786 * \frac{80^2}{380} = \%7,41 < \%8$$

No need for speed limit.

R > 200 m no need for expansion.

$$h_1 = q * \frac{b}{2} = 0,02 * \frac{11}{2} = 0,11m \qquad h_2 = q * \frac{b}{2} = 0,0741 * \frac{11}{2} = 0,408m$$

$$K = \frac{2 * h_1 * L_g}{h_2 - h_1} \Rightarrow 43,71 = \frac{2 * 0,11 * L_g}{0,408 - 0,11} \Rightarrow L_g = 59,21m$$

Clothoid length should be checked for 3 conditions and 3 controls.

Calculation of clothoid parameter

$$L_g = \frac{A^2}{R} \Rightarrow A = \sqrt{R * L_g} = \sqrt{380 * 59,21} = 150m$$

Optical condition

$$\frac{R}{3} \le A \le R \Rightarrow \frac{380}{3} \le 150 \le 380$$

Dynamic condition

$$A_{\min} = 0,19 * \sqrt{V_p^3} = 0,19 * \sqrt{80^3} = 136 < 150$$

There is no need for superelevation condition!

Minimum length control

59,21 m > 45 m

Geometrical control

$$D_2 = \frac{2\pi R_2 \Delta_2}{360} = \frac{2\pi * 380 * 31,83}{360} = 211,1m$$

$$\frac{L_g}{2} \le \frac{D}{2} \rightarrow \frac{59,21}{2} \le \frac{211,1}{2}$$

Jerk control

$$p' = \frac{V_p^3}{46,7 * R * L_g}$$

$$= \frac{80^3}{46,7 * 380 * 59,21} = 0,49m/s^3 < 0,6m/s^3$$

Question – 2

A highway with (2x2) lanes and its geometrical data and the project criteria are given below. Design speed of the road 80 km/h and 44% of the centrifugal force will be compensated by friction force.

Cross-Section No	Km
Φ_9	3+150
45	3+175
$\digamma_9$	3+650

Project Criteria

Platform width = 10 m, Max superelevation = %8

Crown slope = %2, $q_{theo} = 0.786*(V_p^2 / R)$

$p' = V^3 / (46.7*R*L) < 0.6$ m/sn³, $A_{min} = 0.19*V_p^{(3/2)}$

a) The curve angle is given as $\Delta_9 = (71.62)°$. Find all other components of the horizontal curve (D_9=?, R_9=?, t_9=?, b_9=?).

b) On this curve find the length and parameter of the transition curve with largest possible clothoid parameter A. Draw constant centerline superelevation diagram without ramp and calculate necessary values.

c) For the cross-section no (45) calculate superelevation values for inner and outer edges.

a) The curve angle is given as Δ_9= (71.62)°. Find all other components of the horizontal curve (D_9=?, R_9=?, t_9=?, b_9=?).

$$D_9 = \text{Ŧ}_9(km) - \text{Ф}_9(km) = (3+650) - (3+150) = 500m$$

$$D = \frac{2\pi R\Delta}{360} \Rightarrow 500 = \frac{2\pi R_9 * 71,62}{360} \Rightarrow R_9 = 400m$$

$$t_9 = R_9 * tg(\frac{\Delta}{2}) = 400 * tg(\frac{71,62}{2}) = 288,59m$$

$$b_9 = R_9 * (\sec(\frac{\Delta}{2}) - 1) = 93,24m$$

b) On this curve find the length and parameter of the transition curve with largest possible clothoid parameter A. Draw constant centerline superelevation diagram without ramp and calculate necessary values.

$$q_{pra} = 0,56 * q_{teo} = 0,56 * 0,786 * \frac{V_p^2}{R} = 0,56 * 0,786 * \frac{80^2}{400} = 7,04 < 8$$

No need for speed limit.

R > 200 m no need for expansion.

Optical condition

$$\frac{R}{3} \le A \le R \Rightarrow \frac{400}{3} \le A \le 400$$

It is asked for maximum possible A value, so A = 400.

$$L_g = \frac{A^2}{R} = \frac{400^2}{400} = 400m$$

Dynamic condition

$$A_{\min} = 0,19 * \sqrt{V_p^3} = 0,19 * \sqrt{80^3} = 135,95m$$

There is no need for superelevation condition!

Minimum length control

$L_g = 400\ \text{m} > 45\ \text{m}$

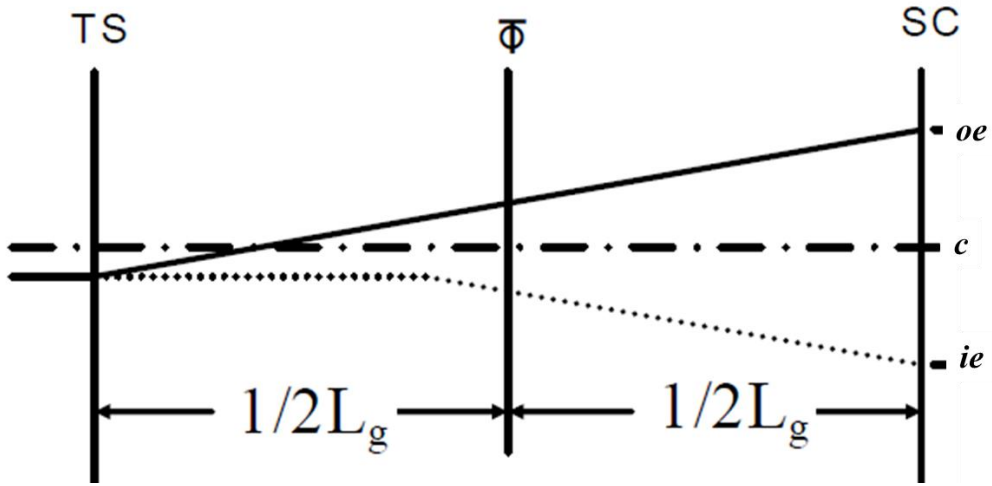
Geometrical control

$$\frac{L_g}{2} \le \frac{D}{2} \rightarrow \frac{400}{2} \le \frac{500}{2}$$

Jerk control

$$p' = \frac{V_p^3}{46,7 * R * L_g}$$

$$= \frac{80^3}{46,7 * 400 * 400} = 0,069 m/s^3 \quad < 0,6 m/s^3$$

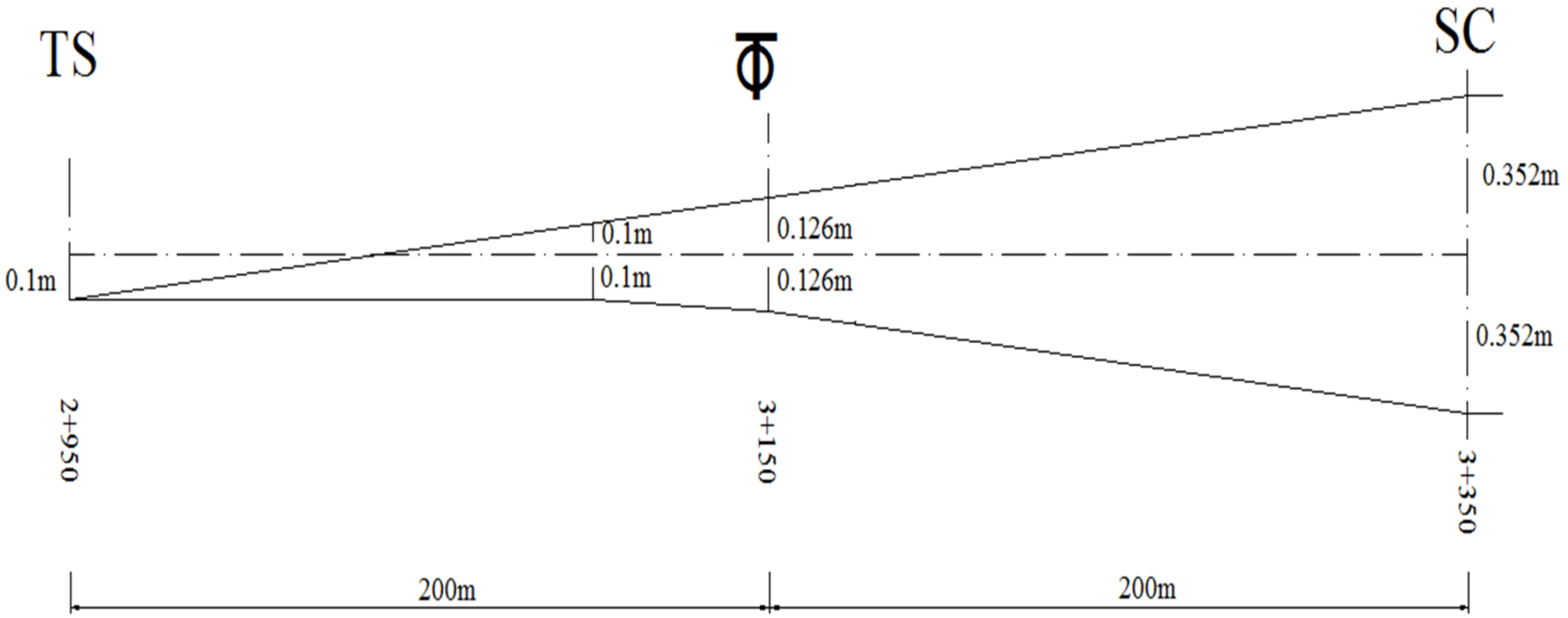


$$h_{TS} = q * \frac{b}{2} = 0,02 * \frac{10}{2} = 0,1m$$

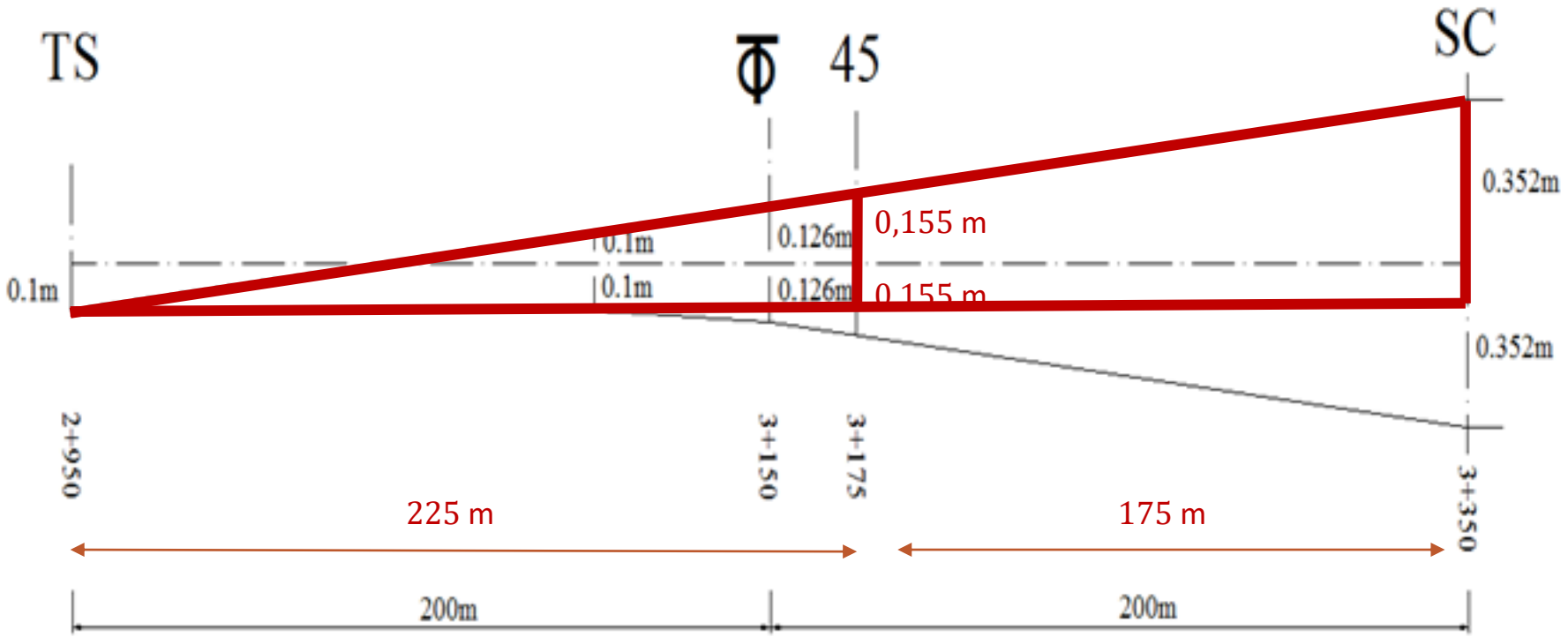
$$h_{SC} = q * \frac{b}{2} = 0,0704 * \frac{10}{2} = 0,352m$$

$$q_{\Phi} \rightarrow \frac{x+2}{7,04+2} = \frac{1}{2} \Rightarrow q_{\Phi} = \%2.52$$

$$h_{\Phi} = 0,0252 \frac{10}{2} = 0.126\ m$$


TS
Φ
SC
0.352m
0.1m
0.126m
0.1m
0.1m
0.126m
0.352m
2+950
3+150
3+350
200m
200m

c) For the cross-section no (45) calculate superelevation values for inner and outer edges.



$$q_{45} \rightarrow \frac{x+2}{7{,}04+2} = \frac{225}{400} \qquad \Rightarrow q_{45} = \%3{,}09$$

$$h_{45} = 0{,}0309\,\frac{10}{2} = 0{,}155\ m$$

Question – 3

A state highway with two directions and double lane is designed for 80 km/h project velocity. The data about the highway and some other parameters are provided below.

Cross-Section	Km	Curve Component
Φ	1+ 527	Δ= (47,75)º
Ғ	1+ 777	

Crown slope = % 2

Lane width = 3.5 m

Shoulder width = 3.0 m

Teoretical Superelevation = $0.786 * (V^2 / R)$

Jerk: $p' = V^3 / (46.7 * R * L) < 0.6\ m/s^3$

Ramp Length: $K = (2 * h_1 * L) / (h_2 - h_1)$

a) It is required that the maximum superelevation should be 8%. At most how much of the centrifugal force can be compensated by superelevation?

b) Find the length and parameter of the smallest possible clothoid which can be placed in to this curve.

c) By taking A as 150 m, it is required to implement constant centerline superelevation with ramp along the clothoid. According to that draw superelevation diagram by making necessary measurements and write Km values for critical points.

a) It is required that the maximum superelevation should be 8%. At most how much of the centrifugal force can be compensated by superelevation?

$$q_{pra} = k * q_{teo} \quad = k * 0,786 * \frac{V_p^2}{R} = k * 0,786 * \frac{80^2}{300} = 8 \qquad \Rightarrow k = 0,477$$

That means 47.7% of centrifugal force can be compensated by superelevation.

b) Find the length and parameter of the smallest possible clothoid which can be placed in to this curve.

In order to calculate the clothoid components, the necessary parameters are; the superelevation radius of curve and the maximum allowable speed on the curve. If there is no speed limitation the maximum speed is equal to the project speed

$$D = \text{Ғ}_{km} - \text{Ф}_{km} = (1+777) - (1+527) = 250m$$

$$D = \frac{2\pi R\Delta}{360} \Rightarrow 250 = \frac{2\pi R * 47,75}{360} \Rightarrow R_1 = 300m$$

No need for speed limit. ***q = %8***

R > 200 m no need for expansion.

Optical condition

$$\frac{R}{3} \le A \le R \quad \Rightarrow \frac{300}{3} \le A \le 300$$

Dynamic condition

$$A_{\min} = 0,19 * \sqrt{V_p^3} \quad = 0,19 * \sqrt{80^3} \quad = 135,95m$$

There is no need for superelevation condition!

It is asked for minimum possible A value, so A is 140.

$$L_g = \frac{A^2}{R} \quad = \frac{140^2}{300} = 65,33m$$

Minimum length control

L_g = 65,33 m > 45 m

Geometrical control

$$\frac{L_g}{2} \le \frac{D}{2} \rightarrow \frac{65,33}{2} \le \frac{250}{2}$$

Jerk control

$$p' = \frac{V_p^3}{46,7 * R * L_g}$$

$$= \frac{80^3}{46,7 * 300 * 65,33} = 0,56m/s^3 < 0,6m/s^3$$

c) By taking A as 150 m, it is required to implement constant centerline superelevation with ramp along the clothoid. According to that draw superelevation diagram by making necessary measurements and write km values for critical points.

Optical condition

$$\frac{R}{3} \le A \le R \quad \Rightarrow \frac{300}{3} \le 150 \le 300$$

Dynamic condition

$$A_{\min} = 0,19 * \sqrt{V_p^3} = 0,19 * \sqrt{80^3} = 135,95m < 150m$$

There is no need for superelevation condition!

Clothoid length:

$$L_g = \frac{A^2}{R} = \frac{150^2}{300} = 75m$$

Minimum length control

L_g = 75 m > 45 m

Geometrical control

$$\frac{L_g}{2} \le \frac{D}{2} \rightarrow \frac{75}{2} \le \frac{250}{2}$$

Jerk control

$$p' = \frac{V_p^3}{46,7 * R * L_g}$$

$$= \frac{80^3}{46,7 * 300 * 75} = 0,487m/s^3 < 0,6m/s^3$$

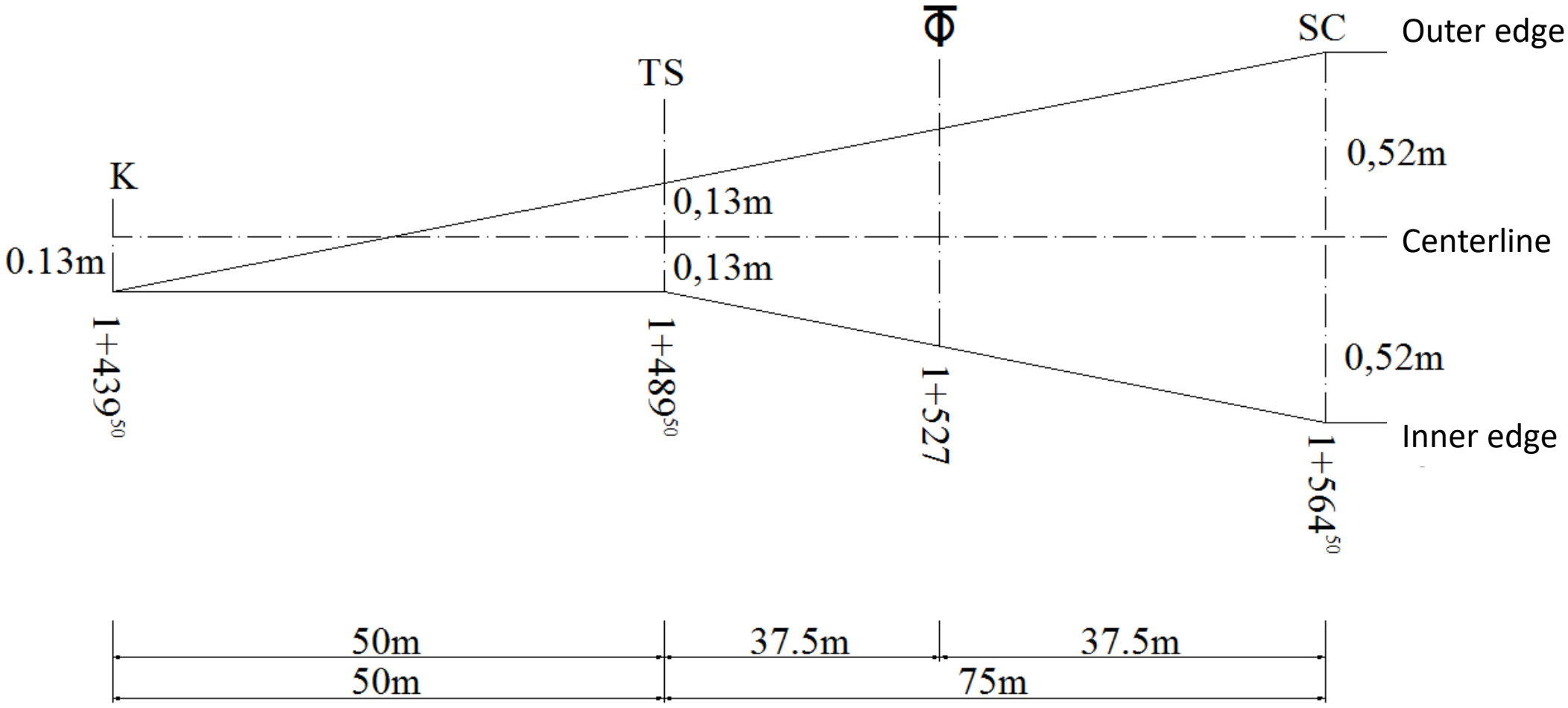
Ramp formula: Need for calculation of h_1 and h_2 ; we know L_g = 75m.

$$K = \frac{2 * h_1 * L_g}{h_2 - h_1}$$

b = 3,5*2 + 3*2 = 13 m

$$h_1 = q * \frac{b}{2} = 0,02 * \frac{13}{2} = 0,13m \qquad h_2 = q * \frac{b}{2} = 0,08 * \frac{13}{2} = 0,52m$$

$$K = \frac{2 * h_1 * L_g}{h_2 - h_1} = \frac{2 * 0,13 * 75}{0,52 - 0,13} = 50m$$



$$q_{\phi} \rightarrow \frac{x+2}{8+2} = \frac{87{,}5}{125} \Rightarrow q_{\phi} = \%5$$

$$h_{\phi} = 0{,}05\frac{13}{2} = 0{,}325\ m$$